

Product Backlog

Backlog ID	Epic/Story Title	User Story (if applicable)	Priority	Story Points	Acceptance Criteria Example	Sprint Target	Dependencies	Risks & Mitigations
EPIC-1	Existing Software Foundation	As a developer, I want to access and review the existing Python-based image-guided intervention software, so that I can understand its architecture and performance.	High	13	Given access to the existing software, When the system is run with provided data, Then the setup completes successfully and architecture components are documented.	Sprint 1	Access to existing repo and data	Missing dependencies → add environment setup scripts
STORY-1.1	Setup and Documentation	As a developer, I want to set up the existing software in a controlled environment, so that I can run and test it.	High	5	Given a configured environment, When setup scripts are executed, Then the software installs without errors and documentation describes setup steps clearly.	Sprint 1	EPIC-1	Environment conflicts → use Docker or virtualenv fallback
STORY-1.2	Database Setup for System Data	As a developer, I want a basic database established, so that the system data can be securely stored and accessed consistently.	High	5	Given the identified data requirements, When the database schema and connection layer are created, Then the system can read and write required data clearly.	Sprint 1	EPIC-1	Misalignment with workflow → keep schema minimal and revise as the requirements evolve
STORY-1.3	UI / UX Update	As a user, I want the UI reviewed and updated for basic usability, so that the existing workflow is clearer and easier to navigate.	Medium	3	Given the current interface, When usability issues are addressed, Then the UI presents clearer navigation, labels, and layout for the workflow steps.	Sprint 1	EPIC-1	Inconsistent updates → follow existing workflow structure and apply targeted

								changes
EPIC-2	Modular Architecture	As a developer, I want the architecture to be modular, secure, and extensible, so that future enhancements can be efficiently integrated.	High	13	Given the existing architecture, When modules are refactored, Then the interfaces are clear and code dependencies are minimized.	Sprint 1	EPIC-1 complete	Integration bugs → introduce staged merges and validation
STORY-2.1	Refactor Modules	As a developer, I want to refactor the code into independent modules, so the system is easier to extend.	High	8	Given existing functions, When modules are separated, Then each compiles independently and retains the same functionality.	Sprint 1	EPIC-2	Modules don't connect properly → define clear API contracts
STORY-2.2	Implement Security Foundations	As a developer, I want basic security practices integrated with the existing database and system operations, so that they are protected.	High	5	Given the modular architecture and database, When security measures are applied, Then the system ensures safely handling sensitive data.	Sprint 2	Database Setup (STORY-1.2), EPIC-2	Inconsistent security practices → document and apply security steps across all modules
STORY-2.3	System Availability	As a stakeholder, I want basic availability measures in place, so that the system remains stable and runs without unexpected interruption.	High	3	Given the refactored system, When availability measures are applied, Then the system is consistent during typical usage and handles failure gracefully.	Sprint 2	EPIC-2	Undefined reliability measures → establish baseline error handling patterns for all modules.

EPIC-3	2D Images to 3D Volume Reconstruction	As a surgeon, I want accurate 3D reconstruction from 2D ultrasound images, so that I am able to make better surgical decisions.	High	32	Given a set of 2D ultrasound images, When the reconstruction pipeline runs, Then a 3D volume is produced within or improved reference accuracy.	Sprint 2	Modular architecture (EPIC-2)	Slow performance → use profiling to guide optimization
STORY-3.1	Analyze Current Code (Python and C++)	As a developer, I want to thoroughly analyze and review the existing reconstruction code, so that I can identify inefficiencies.	High	5	Given current reconstruction scripts, When profiling tools are executed, Then bottlenecks and memory inefficiencies are logged and documented.	Sprint 2	EPIC-3	Performance issues not obvious → use profiling to identify and focus on the slowest code sections
STORY-3.2	Prototype Optimization	As a developer, I want to prototype an optimized reconstruction pipeline, so that performance is improved.	High	8	Given identified bottlenecks, When optimization changes are applied, Then the system runs faster while maintaining at least equivalent 3D results.	Sprint 2	STORY-3.1	Optimizations could reduce accuracy → validate each change against baseline reconstruction outputs before merging
STORY-3.3	Tracked Ultrasound Sequence Generation (.mha)	As a developer, I want ultrasound frames and tracking data exported as a PLUS-compatible .mha file, so that the reconstruction and calibration tools can consume the data directly.	High	8	Given captured ultrasound frames and tracking transforms, When a capture sequence is completed, Then a valid .igs.mha sequence metafile is generated.	Sprint 2	EPIC-2 Modular Architecture	Incorrect format prevents calibration → validate before use

STORY-3.4	Calibration Configuration Integration	As a developer, I want to generate configuration files for calibration tools, so that the system can automatically run temporal calibration and reconstruction.	Medium	3	Given a captured tracked sequence, When calibration is initiated, Then the configuration file launches PLUS calibration successfully.	Sprint 2	EPIC-2 Modular Architecture	Incorrect parameters prevent calibration → validate configuration against sample dataset
STORY-3.5	Reconstruction Integration	As a surgeon, I want tracked ultrasound data reconstructed into a 3D volume using the PLUS reconstruction pipeline, so that I can visualize anatomical structures in three dimensions during procedures.	High	8	Given a calibrated tracked sequence, When reconstruction is triggered, Then a 3D ultrasound volume is generated and the volume is spatially consistent with tracking data.	Sprint 2	STORY-3.3	Incorrect calibration prevents reconstruction → validate spatial and temporal calibration before reconstruction